

Leveraging Nash Equilibrium and Generative Adversarial Networks for Autonomous Driving

Abstract:

Autonomous driving in the context of Advanced Driver Assistance Systems (ADAS) and Highly Automated Driving (HAD) has evolved over the last couple of decades with an increasing significance towards Deep Learning principles from Artificial Intelligence. By combining this development with Nash Equilibrium, a concept from game theory, this paper discusses how we could establish and compute ground truths from vehicular videos used to train, test, and validate autonomous driving sensors and systems.

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Introduction

Artificial Intelligence (AI) in ADAS, HAD, and Autonomous Driving (AD) has become significant in recent times. This has led to the widespread practice of ascertaining the party's role(s) and responsibilities genuinely responsible for causing a road accident. This list comprises the vehicle's Original Equipment Manufacturer (OEM), its supplier be it Tier -1 or Tier -2, AI algorithms, the end-user, or all parties being equally responsible. Therefore, [ascertaining the cause of an accident is, at this point, difficult or not possible at all](#).

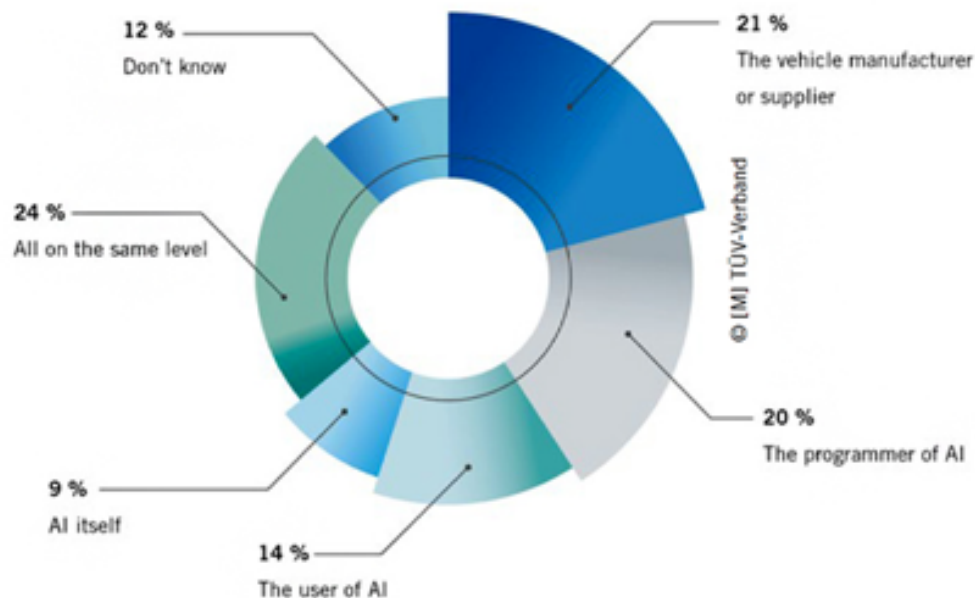


Figure 1. Factors responsible for road accidents | Source: [Marklines](#)

At the start of the 21st century, only defense agencies such as Defense Advanced Research Projects Agency (DARPA) and select military intelligence worldwide were attempting unmanned vehicles to explore unknown territories. Slowly the use of sensors such as cameras and ultrasonic sensors started gaining traction with OEMs and suppliers. Recent advancements have also provided access to additional sensors such as radar, lidar, and High Definition (HD) maps.

These architectures were simple initially, focusing on pixel-by-pixel or frame-based processing as an example in terms of camera algorithms. Gradually, they became more complex to handle and lacked architecture that supported the necessary speed. Graphic Processing Units (GPUs) such as those made by Nvidia entering the market led to increased use of Artificial Intelligence (AI) and Deep Learning (DL), including TensorFlow (TF) from Google.

Game theory comprises many techniques, including Nash Equilibrium, which could play a significant role in payoff strategies for determining AD systems' ground truth. This paper explains how Nash Equilibrium and DL can be combined to provide highly reliable systems and ascertain system benchmarking with ground truth realities to a certain degree.



Components of AD Architecture

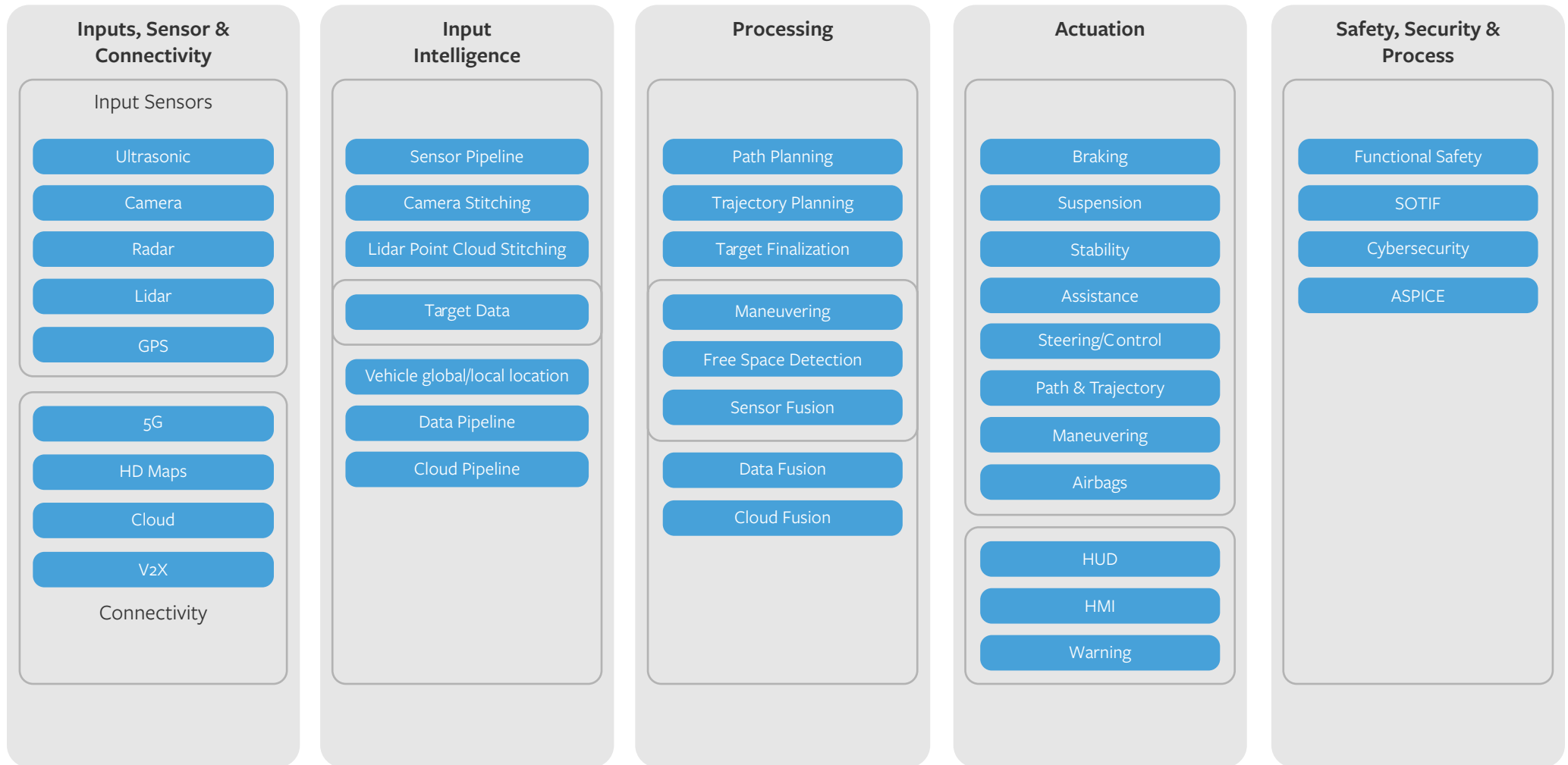


Figure 2. Components of AD Architecture



Inputs, Sensors & Connectivity: For any Autonomous Driving (AD) system, the perception of the self or the ego vehicle at the current location is pertinent to determine the safe and secure path it takes in the future. This is managed by sensors such as ultrasonic sensors, cameras, radar, lidars, and Global Positioning System (GPS). This, along with connectivity enabled features such as 4G/5G, HD maps, Vehicle to Everything (V2X), and Cloud connectivity to back-end telematics units, enables live updates of vehicular position, location, direction, and other parameters.

Input Intelligence: The back-end of input sensors and connectivity is led into pipelines for pre-processing inputs with the AD vehicle's required processing units. Some examples include sensors, data, and cloud pipelines. Specific aspects like camera stitching and lidar point cloud stitching show the ego vehicle's position relative to the target data around it. GPS, telematics, V2X, and Cloud provide the vehicle with location details to enable precision-based routing details in HD Map-based outputs.

Processing: Most algorithms to determine what needs to be done with the processed inputs are in Electronic Control Units (ECUs), Domain Control Units (DCUs), and Zone Control Units (ZCUs), usually connected to a central Vehicle Computing Units (VCUs). These ADAS DCUs determine most of the sensor, data, and cloud fusion to determine path planning, trajectory planning, target finalization, maneuvering, and free space detection. These aspects are required to provide specific alarms or actuating to make the AD vehicle move with or without driver intervention.

Actuation: Outputs to AD vehicle with or without driver intervention are provided via warnings, Human Machine Interface (HMI), Heads-Up Display (HUD), or by actuators like braking, suspension, stability, airbags, steering, or control. This will maneuver the AD vehicle without any collision and making them seamlessly handle any scenario.

Safety, Security, and Process: All the subsystems, sensors, actuators, and processing units need to comply with safety and security standards as determined by Vehicle Regulatory Authorities and process compliances. These would be defined by the organization and production policies in all geographies, as non-compliance would lead to untoward incidents resulting in recalls.



Game Theory – Nash Equilibrium Benefits

Nash equilibrium provides quantitative predictability of winners in a multi-player cooperative or non-cooperative scenario. Its usage would make it easier to interpret and exchange the roles of [parties involved in AD vehicles in any given scenario with determined predictability](#).

		Mac	
John		Confess	Remain Silent
Confess		5.5	2.10
Remain Silent		10.2	0.0

Figure 3. Payoff Matrix | Source: [EconomicsDiscussion.net](https://www.economicsdiscussion.net)

A simple example of its application is in a scenario known as the prisoner's dilemma. The prisoner's dilemma is a problem that can be emulated for two prisoners. If both confess, they might serve a life sentence. If one confesses and the other remains silent, each will get imprisonment based on their prior sentences. If both remain silent, they might receive a minimum term. In this scenario, each possibility can be represented as [a payoff matrix to determine the best possible strategy for a shorter prison sentence](#). Such non-cooperative games provide enough use cases for Nash equilibrium and methods to implement strategies for each prisoner. The winning strategy prefers the best possible probability for all the actors in a non-cooperative scenario. Thus, making use of gaming theory principles, like a simple Nash equilibrium, gives the best possibility of predicting the best outcomes in chaotic scenarios with multiple actors in any environment. It has been used in various fields such as game strategies, economics, strategic management, and numerous business scenarios such as supply-chain management and others.



DL – GAN Benefits

In the context of AI and DL, Generative Adversarial Networks (GANs) have paramount significance in determining the discriminators' multi-variable truth. This leads to a close relationship of the test data on the trained data, [as seen in scenarios with different labeled and prepared datasets](#). This can be closely used to monitor datasets with varying probabilities and scenarios, leading to the best possible outcomes in a computer vision-like ecosystem.

Usually, a confusion matrix gives a list of probabilities of true and false positives for the multiple variables and their forward and backward propagation of the same in a GAN. We can usually identify the best possible outcomes in a set of images and derive the ideal predicted behavior and the ground truth's dominance in the pictures.

As seen in Figure 4, the rightmost column of a) gives the best possible predicted scenario in a handwriting recognition system. The rightmost column of b) shows the best predicted facial expressions. The following tables provide the best predictions of animals as close to the original ones, also known as the 'ground truth' of the trained image datasets.

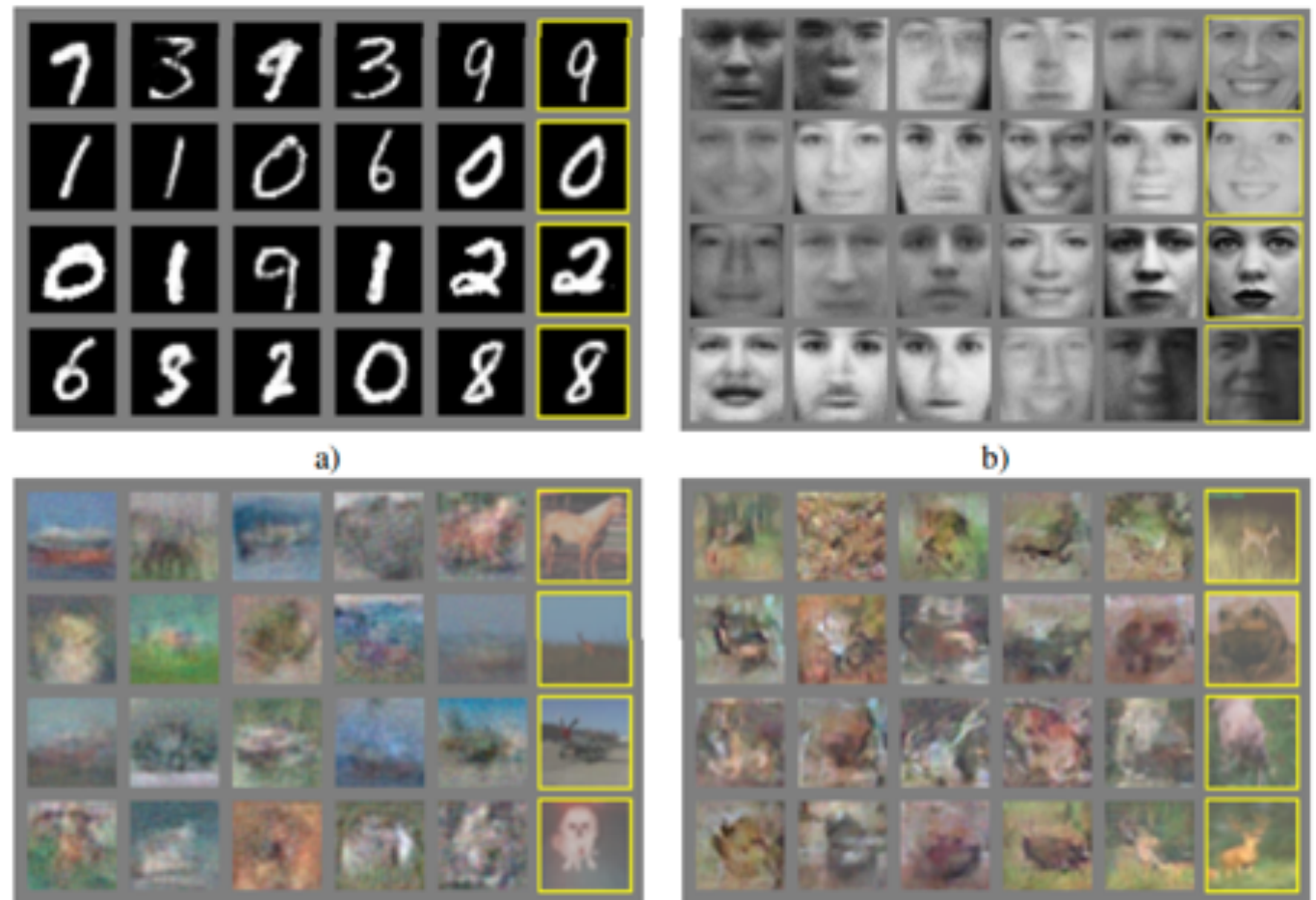


Figure 4. GAN determining the best results for trained datasets | Source: [NeurIPS](#)



Successful Case Studies Leveraging Nash Equilibrium

- Commerce.ai, a company that harnesses AI in designing commercial products, [uses both GAN and Nash Equilibrium for a range of shoe products.](#)
- Universities in China, including the Naval Laboratories, have collaborated to [determine gearboxes' predictive maintenance using GANs and Nash Equilibrium.](#)
- A new approach for protein synthesis has been implemented by Texas A&M University in the USA using Conditional Wasserstein GANs, which also [derives principles from GANs with or without Nash equilibrium.](#)
- A University in Delhi produced a publication that describes [enhanced image restoration by GANs using Game Theory.](#)



Solution/Deployment Scenario - AD Vehicle in a Roundabout

Roundabouts are encountered at several urban and rural roads, many times with no traffic light. This scenario lends itself to several approaches for AD vehicles to handle different conditions to enter and exit the roundabout. This includes following the line of vehicles entering the roundabout and reaching the roundabout. Once there, determining the timing window to enter, gauging intersection speed to enter, merging with other vehicles, calculating other entry points, and waiting for the exit to approach and follow the maneuver for a seamless exit. Several approaches have been suggested for such scenarios. Developing a collaborative game strategy with Nash equilibrium and GANs gives a winning approach for the best-fit or near-to-ground truth estimation. The confusion matrix taking the probabilities for multivariate scenarios is illustrated in Figure 5.

		AD vehicle											
Ego vehicle		ACTUAL VALUES		PREDICTED VALUES									
		Entering vehicle types	Entering vehicle parameters	Time to reach roundabout	Speed for stoppage	Time for stoppage	Timing window at Roundabout	Intersection speed	Enter window	Merging maneuver	Other entry point vehicle parameters	Exit window time	Exit window maneuver
1	Entering vehicle types	0.712652	0.572905	0.558465737	0.778797	0.990008	0.18859422	0.891083	0.520745	0.746556	0.167875842	0.379299	0.170339
2	Entering vehicle parameters	0.701362	0.934787	0.025717694	0.571139	0.499001	0.34294464	0.462088	0.696708	0.137685	0.171695153	0.832518	0.579444
3	Time to reach roundabout	0.561895	0.152263	0.709477326	0.006087	0.454673	0.88974169	0.488677	0.982897	0.778913	0.337976959	0.344167	0.290617
4	Speed for stoppage	0.675188	0.427988	0.144201246	0.505994	0.664559	0.24819441	0.636794	0.571889	0.250113	0.690733503	0.661683	0.479795
5	Time for stoppage	0.182934	0.966406	0.446709992	0.687182	0.602011	0.49269274	0.56898	0.733797	0.082753	0.36192829	0.581027	0.221816
6	Timing window at Roundabout	0.17673	0.785998	0.818119546	0.159286	0.482296	0.18800209	0.988745	0.793099	0.272732	0.52513589	0.787913	0.93983
7	Intersection speed	0.60545	0.384544	0.825428676	0.548993	0.924944	0.35644232	0.62326	0.86582	0.811069	0.715283889	0.96659	0.82239
8	Enter window	0.444988	0.107683	0.344274121	0.253159	0.276008	0.40819626	0.078833	0.654326	0.191599	0.822575818	0.671921	0.384655
9	Merging maneuver	0.749631	0.530948	0.516944127	0.822209	0.784718	0.30225631	0.704544	0.843269	0.934017	0.491069417	0.601639	0.006496
10	Other entry point vehicle parameters	0.267105	0.171066	0.427781004	0.459602	0.414288	0.78941634	0.656465	0.743237	0.099674	0.748195518	0.411498	0.6001
11	Exit window time	0.7208	0.69752	0.3853081	0.169201	0.131624	0.593201	0.204287	0.656642	0.777971	0.026987622	0.016759	0.783622
12	Exit window maneuver	0.583328	0.589163	0.23370757	0.170687	0.757867	0.91395348	0.15914	0.359512	0.692445	0.001738197	0.346259	0.149652

Figure 5. Confusion Matrix illustrating possible probabilities for a roundabout scenario



Impact of Solution

The confusion matrix probabilities can form a basis for algorithms computing traffic conditions, time to collision, time to merge, target windows, and maneuvering criteria based on GANs and Nash equilibrium. This would provide the best case fit with the ground truth. These can be [obtained in a similar manner, as seen in a prior study](#). Some of the notable illustrations and interpretations close to ground truth based on the confusion matrix in Figure 5 are as follows:

Entering Vehicle Types

At a roundabout, depending on the type of vehicles entering, it has a probability of ~ 0.5 (actual value) to determine the time to reach the roundabout. Due to a variety of AD vehicles, the probability is 50/50 for the various autonomous sensors. The predicted value is also approximately 0.5 and closer to or greater than the actual value considering the system is close to the ground truth based on prediction.

Entering Vehicle Parameters

If there is a lot of traffic in front of the ego vehicle at the roundabout, then vehicle entry parameters will change. Braking, for instance, would increase to intermittent

braking. This is because the probability of time to reach the roundabout is less than around 2 percent or ~ 0.02 (actual value). However, the predicted value of the entering vehicle parameters to probability to reach the roundabout is more than 0.15. Hence, the actual value suggests that the ego vehicle is taking a much cautious approach to the roundabout and is, therefore, closer to the ground truth.

Time to reach the roundabout

In this case, the ego vehicle can sense from a map that there is almost no traffic at the roundabout. The probability of time to reach roundabout to the roundabout's timing window is very high or approximately equal to ~ 0.88 . The predicted value is just 0.81 making the AD vehicle much better at actuals and closer to the ground truth.

Speed for stoppage

Suppose the ego vehicle is a well-maintained vehicle with precision braking and maneuvering capabilities. The probability of speed for a halt to time for the halt is medium, i.e., ~ 0.67 . This keeps in mind the amount of time required to stop and go around the roundabout. The

predicted value is 0.68 making the AD vehicle better at actuals and closer to the ground truth.

Time for stoppage

Suppose we see that the probability of time for a halt is relatively high at ~ 0.73 . This means that whenever the ego vehicle stops at the intersection, there is a relatively higher chance of predicting the entry window into the roundabout and completing the roundabout maneuver. The predicted value is ~ 0.27 , indicating that the learning is much better by the AD vehicle and hence much greater than the actual value and therefore closer to the ground truth.

Timing window at a roundabout

The relatively high probability of ~ 0.99 for timing window at a roundabout to intersection speed shows that the roundabout is well controlled. It also indicates that all other vehicles follow traffic rules and might be well equipped with AD systems or ADAS systems. Like the last scenario, the AD vehicle performed much better at actuals and is closer to the ground truth.



Intersection speed

The relatively high probabilities of intersection speed to enter and exit windows show that there is less traffic, and the AD vehicle can approach the roundabout with a somewhat higher intersection speed. As seen earlier, the AD vehicle performed better at actuals under two parameters and is closer to the ground truth.

Entry window

The highest probability of entry window to other entry point vehicle parameters suggests that the ego vehicle is equipped with many sensors to monitor all other entry points and make decisions at the roundabout. With the predicted value of 0.74 versus the actual value of 0.82, the AD vehicle is closer to the ground truth.

Merging maneuver

The mid probability of ~0.49 for the merging maneuver to other vehicle entry point parameters suggests that the actual value is better than the predicted value of 0.09. It also indicates that the AD vehicle capabilities perform the best in the scenario and are closer to ground truth.

Other entry point vehicle parameters

The low probability of ~0.1 for other entry point vehicle parameters with the merging maneuver might suggest that if other vehicles are around the ego vehicle, entry points are less equipped with AD features. This would be chaotic, and the merging maneuvers would not be easy, making an accident highly likely. However, the actual probability of ~0.49 suggests that the AD vehicle performs better here and is closer to the ground truth, with nearly more than five times the chances of avoiding an accident.

Exit window time

The exit window time will be relatively low if the other entry point vehicle parameters are highly predictable, meaning they are AD-equipped, giving a very low probability of less than 0.03. However, the actual value ~0.41 suggests that the AD vehicle is near the ground truth because of various entry point vehicles' multiple parameters in the roundabout.

Exit window maneuver

The high probability of 0.34 for exit window maneuver to exit window time suggests that more probable is a seamless exit at the roundabout with more time. In this case, too, the actual value is high at ~0.78, suggesting that the AD vehicle is performing closer to the ground truth. From a Nash Equilibrium perspective, the strategic winning probability for the listed 12 tuple scenarios will be based on the multiplication of all the 12 possible probabilities. This would involve multiple vehicles in a non-cooperative state with each other. The same applies to other AD vehicles as the participants can be exchanged in Nash Equilibrium.

The best possible scenario is obtained from a GAN perspective when the actual value is closer or better than the predicted value. This indicates that the AD vehicle performs in line with the ground truth, closer to the mid-line in the confusion matrix.



Conclusion

In this publication, we have provided the means to decide how the algorithm performs based on a confusion matrix. This can be used both in GAN and Nash Equilibrium to predictably point towards the right direction that the AD vehicle is close to the ground truth. However, it is left to the users to conclude based on current limitations, as the action basis for an accident in an AD vehicle using AI is still difficult to ascertain. At this point in time, the discretion of regulatory agencies to demystify the cause is still required.

About the Author

Naresh Neelakantan is currently a Senior Architect in Product Engineering Services at Sasken Technologies Limited. With 16 years of experience working with various Automotive companies, he is currently helping Sasken make strategic decisions in the areas of AUTOSAR and Autonomous Driving while aligning the Automotive ECU-to-DCU market with Sasken's Chip-to-Cognition vision. He is a subject matter expert in Research & Technology Management of Embedded Software Architectures combining Analog sensors with Digital technologies. He has been an integral part of teams that have successfully delivered multiple products which have been implemented by Tier-1s and are a part of global automotive OEMs.

About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to-market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, SatCom, Telecom, and Transportation industries. For over 30 years and with multiple patents, Sasken has transformed the businesses of 100+ Fortune 500 companies, powering more than a billion devices through its services and IP.





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